

DW01_decode

Decoder

Version, STAR, and myDesignWare Subscriptions: [IP Directory](#)

Features and Benefits

- Parameterized word length
- Inferable using a function call

Revision History

Description

DW01_decode decodes an address on input port A to a single bitline on output port B. The selected bitline on port B is active high.

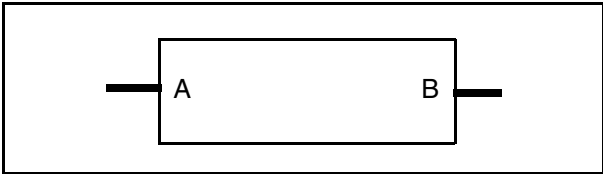


Table 1-1 Pin Description

Pin Name	Width	Direction	Function
A	<i>width</i>	Input	Binary input data
B	2^{width}	Output	Decoded output data

Table 1-2 Parameter Description

Parameter	Values	Description
<i>width</i> ^a	≥ 1	Word length of input A is <i>width</i> Word length of output B is 2^{width}

a. The *width* parameter value causes the size of output B to grow exponentially. Therefore, a *width* value greater than 12 results in abnormally long compile times.

Table 1-3 Synthesis Implementations

Implementation Name	Function	License Feature Required
str	Synthesis model	none

Table 1-4 Simulation Models

Model	Function
DW01.DW01_DECODE_CFG_SIM	Design unit name for VHDL simulation
dw/dw01/src/DW01_decode_sim.vhd	VHDL simulation model source code
dw/sim_ver/DW01_decode.v	Verilog simulation model source code

Table 1-5 Truth Table (width = 3)

A(2:0)	B(7)	B(6)	B(5)	B(4)	B(3)	B(2)	B(1)	B(0)
000	0	0	0	0	0	0	0	1
001	0	0	0	0	0	0	1	0
010	0	0	0	0	0	1	0	0
011	0	0	0	0	1	0	0	0
100	0	0	0	1	0	0	0	0
101	0	0	1	0	0	0	0	0
110	0	1	0	0	0	0	0	0
111	1	0	0	0	0	0	0	0

Related Topics

- [Logic – Combinational Overview](#)
- [DesignWare Building Block IP User Guide](#)

HDL Usage Through Function Inferencing - VHDL

```
library IEEE, DWARE;
use IEEE.std_logic_1164.all;
use IEEE.std_logic_arith.all;
use DWARE.DW_foundation_arith.all;

entity DW01_decode_func is
    generic(func_width : integer := 5);
    port( func_A:      in  std_logic_vector(func_width-1 downto 0);
          B_func_TC:  out std_logic_vector(2**func_width-1 downto 0);
          B_func_UNSS: out std_logic_vector(2**func_width-1 downto 0);
          B_func:      out std_logic_vector(2**func_width-1 downto 0));
end DW01_decode_func;

architecture func of DW01_decode_func is
begin

    B_func_TC  <= std_logic_vector(DWF_decode (signed (func_A)));
    B_func_UNSS <= std_logic_vector(DWF_decode (unsigned (func_A)));
    B_func     <= DWF_decode (func_A);
end func;
```

HDL Usage Through Function Inferencing - Verilog

```
module DW01_decode_func (func_A,B_func);  
    parameter func_width = 4;  
  
    // Passes the width to the decode function  
    parameter width = func_width;  
  
    // Please add search_path = search_path + {synopsys_root + "/dw/sim_ver"}  
    // to your .synopsys_dc.setup file (for synthesis) and add  
    // +incdir+$SYNOPSYS/dw/sim_ver+ to your verilog simulator command line  
    // (for simulation).  
    `include "DW01_decode_function.inc"  
  
    input [func_width-1:0] func_A;  
  
    output [(1 << func_width)-1:0] B_func;  
  
    assign B_func = DWF_decode(func_A);  
  
endmodule
```

HDL Usage Through Component Instantiation - VHDL

```
library IEEE,DWARE;
use IEEE.std_logic_1164.all;
use DWARE.DWpackages.all;
use DWARE.DW_foundation_comp.all;

entity DW01_decode_inst is
  generic (inst_width : NATURAL := 8);
  port (inst_A : in std_logic_vector(inst_width-1 downto 0);
        B_inst : out std_logic_vector(2**inst_width-1 downto 0));
end DW01_decode_inst;

architecture inst of DW01_decode_inst is
begin

  -- Instance of DW01_decode
  U1 : DW01_decode
    generic map ( width => inst_width )
    port map ( A => inst_A, B => B_inst );
end inst;

-- pragma translate_off
configuration DW01_decode_inst_cfg_inst of DW01_decode_inst is
  for inst
    end for; -- inst
end DW01_decode_inst_cfg_inst;
-- pragma translate_on
```

HDL Usage Through Component Instantiation - Verilog

Because Verilog does not support an exponentiation operator, you must explicitly set your input and output port widths. Check [Table 1-1](#) on page 1 for details on the relationship between input and output port widths. You also must explicitly set a value for the parameter in your instantiation statement.

```
module DW01_decode_inst( inst_A, B_inst );

    parameter width = 8;

    input [width-1 : 0] inst_A;
    output [(1<<width)-1 : 0] B_inst;

    // Instance of DW01_decode
    DW01_decode #(width)
        U1 ( .A(inst_A), .B(B_inst) );

endmodule
```

Revision History

For notes about this release, see the [DesignWare Building Block IP Release Notes](#).

For lists of both known and fixed issues for this component, refer to the [STAR report](#).

For a version of this datasheet with visible change bars, click [here](#).

Date	Release	Updates
September 2024	DWBB_202409.0	■ Updated date and version; no technical changes
December 2023	DWBB_202312.0	■ Updated date and version; no technical changes
January 2019	DWBB_201806.5	■ Updated example in “ HDL Usage Through Component Instantiation - VHDL ” on page 5 ■ Added this Revision History table and the document links on this page

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